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AI-Enabled Smart Traffic Management for Real-Time Congestion Risk Prediction

Vishnupriya K[1]
Department of Artificial Intelligence and Data Science
Karpagam College of Engineering
Coimbatore, Tamil Nadu, India
vishnupriyaak98@gmail.com

Sanasya S[2]
Department of Artificial Intelligence and Data Science
Karpagam College of Engineering
Coimbatore, Tamil Nadu, India
sanasya04@gmail.com

Maitharayan R[3]
Department of Artificial Intelligence and Data Science
Karpagam College of Engineering
Coimbatore, Tamil Nadu, India
maithrayan8@gmail.com

Sridhar V[4]
Department of Artificial Intelligence and Data Science
Karpagam College of Engineering
Coimbatore, Tamil Nadu, India
ssridharv765@gmail.com

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***Abstract*—Urban traffic jam is one of the most important issues that influence transport efficiency, travel time, consumption of fuel and environmental pollution. The congestion risk prediction and optimized route recommendation system based on Artificial Intelligence presented in this paper suggests a Smart Traffic Management System. The traffic system combines both real-time traffic information available through third-party APIs and past traffic information to record the major variables including the speed of the vehicle, the state of the road, and the time-dependent traffic trends. Random Forest classifier can predict the level of congestion (Low, Moderate, and High) and allows a dynamic route planning, which gives a list of possible options on an interactive map. The users are directed through the path that has the least danger of congestion and the least time. This framework is proactive as opposed to traditional reactive traffic control, which predicts congestion and involves such predictions in routing decisions. We test the system in extreme conditions such as accidents, closure of roads, and events of the population and conduct sensitivity analysis to determine how a classification error affects the process of optimization of routes. Real world experimental results based on actual urban traffic datasets indicate that there is a substantial decrease in the time of transportation and proper prediction of congestion. The paper identifies the applicability of the system to intelligent urban mobility and proposes possible additions to the system by reinforcement learning to further refine the long-term traffic movement.**

Keywords—Traffic congestion, Real-time prediction, machine learning, random forest, Route optimization, Urban mobility, Intelligent transportation.

I. INTRODUCTION

The city traffic jam has become one of the biggest problems of the contemporary city with the rush urbanization, population increase and the number of vehicles. As well as increasing the time of traveling, congestion also results in increased fuel use, environmental pollution, and economic decrease, both to the individual commuter and the general efficiency of transportation in the city. The correct prediction of traffic congestion and optimal route planning are, thus, important elements of intelligent urban traffic control systems. Current research has delved in the area of state-of-the-art machine learning and data-driven solutions to enhance the prediction of congestion. Sravani et al. [1] have suggested a Multilevel-Gated Recurrent Unit (M-GRU) model, which is able to predict traffic congestion more precisely because it is able to adapt to the changes in the road network using features like number of vehicles and weather conditions in comparison to the traditional CNN and Bi-LSTM models. Xu and Ding [2] trained a multi-source fusion model Bi-LSTM with XGBoost in adaptation weighted fusion to traffic congestion prediction to maximize performance levels in RMSE, MAE and accuracy of congestion level identification. Neeraja et al. [3] proposed a cause-aware multi-task learning system that predicts congestion severity, the root causes (e.g., incidents, weather, roadwork) and a Congestion Cost Index, which can be used in action-driven urban mobility planning. Simultaneously, route optimization is a complement to the effect of decreasing the impact of congestion. Zhou and Zhang [4] explored multimodal transportation networks with the help of graph theory and Dijkstra algorithm in an attempt to reduce the overall cost of transportation by enhancing efficiency. Following the strategy, Zhou et al. [5] set forth a data-driven multi-objective optimization framework of multimodal routes under uncertainty, which combined Monte Carlo simulations with evolutionary algorithms to trade-off conflicting goals and offer decision support.

Based on these, the intelligent urban traffic systems should combine real-time congestion prediction with route optimization. Predictive models allow the commuters to use the less congested routes and therefore save time and increase the efficiency of roadways, in addition to improving the mobility of the entire urban region by integrating live traffic information with historical trends.

II. RELATED WORKS

The prediction and management of traffic congestion has been an area of broad research, and techniques have been developed as far as machine learning and combination of optimization and intelligent route planning. A hybrid model of a better particle swarm optimization (IPSO)-based radial basis function (RBF) with LSTM/SVM feature fusion was suggested by Zhang et al. [6] to predict the state of the urban traffic. Not only does their work enhance the accuracy of prediction, but also adds a congestion section control in terms of allocation of traffic validated through VISSIM simulation and showing better performance compared to the traditional control schemes. Shi [7] attempted to discuss the disadvantages of single-structured prediction models by proposing a hybrid model between ARIMA and grey relational analysis model. It is a hybrid method, involving time series analysis and gray system theory to capture the dynamics

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of the traffic flow in an accurate manner to reduce the error in predicting the traffic flow due to environmental factors and offer dependable decision support on the management of traffic flow in a city.

Graph-based deep learning has also turned out to be an opportunity in congestion prediction. A Dual-View Graph Convolutional Neural Network (DV-GCNN) by Jiang and Zhang [8] was proposed in which both spatial and time-related dependencies on traffic data are modeled. The model is effective in predicting congestion levels due to the added congestion indices on the adjacency matrix and the ability to account for the temporal variations based on adjacent and periodic views, which is better than the baseline methods in experimental tests. Sridevi et al. [9] have examined the use of AI to detect congestion and optimize routes in the next-generation smart cities, creating an AI-driven system based on the use of the YOLO-based vehicle detection system, graph-based path finding, and 6G-based low latency networks. The system actively re-routes traffic, provides greater movement of urban areas and decreases the travelling period, carbon dioxide emissions, and fuel expenses, which demonstrate the opportunities of integrating deep learning with sophisticated communication technologies. Subash et al. [10] worked on the optimization of a route based on the integration of road safety and environment as part of the Dijkstra algorithm. Their multi-factor optimization takes into account distance, time of traveling as well as the rate of accidents thus allowing the users to choose safer and efficient paths in real time. The methodology can fill the gaps in the current systems lacking customization and real time flexibility and underlines the practical value of considering safety and sustainability in the traffic management of cities.

The multi objective and metaheuristic models have been used to address the optimization of public transportation and traffic flow. Ma et al. [11] have created a detailed model of the traffic flow of Baltimore City that incorporates all the public transportation nodes, routes and flow volumes in the model. To improve the urban mobility using traffic modeling and optimization techniques, a multi-objective optimization model and genetic algorithm were used to eliminate redundant traffic nodes by 20, but ensured that the travel demands were not compromised, making the reform significant. More sophisticated machine learning models have also helped improve congestion prediction especially in the highways and traffic affected by events. Qian et al. [12] suggested a Multi-task iTransformer model to predict short-term highway congestion which takes into consideration incidents which happen randomly like closing lanes. The architecture combines inverted embeddings and multi-channel attention to calculate traffic flow and category changes caused by events in the environment, and it is accurate by 35.2 percent better than single-task models.

The entropy-based techniques have been applied to the models of congestion and meta-parameter analysis. The model proposed by Mehdi et al. [13] is the differential-entropy approach to labeling traffic states with the subsequent CNNs to predict traffic congestion on the condition of vehicle speed, node position, and special road conditions. The model has been tested and validated on the CityPulse dataset and could precisely predict congestion hotspots, which can be used to re-orient routes to alleviate traffic congestion. Naveen et al. [14] have developed and tested the concept of route optimization with AI and urban commuters by presenting RouteRover, a system that forecasts congestion and displays traffic trends based on machine learning and deep learning. The system involves feature extraction, exploratory data analysis, and manifold learning to detect important traffic elements, which help predict a congestion and optimize the flow, especially in a highly populated city environment. Lastly, Singha et al. [15] surveyed the different AI and deep learning models used to predict traffic congestion. Their research focuses on the performance of neural network structures in the representation of complex traffic dynamics and gives a comparative outlook on various forecasting techniques, which offers an indication of how to choose the right models when forecasting congestion on a short-term and real-time basis.

All these studies reveal how traditional models of traffic modeling have evolved into new advanced AI-based frameworks of prediction and optimization, with a trend showing the adoption of integrated, real-time traffic management solutions which are a combination of the prediction, route optimization and congestion reduction strategies.

III. Proposed System

The Intelligent Urban Traffic Management System suggested is expected to anticipate congestion immediately and suggest the most efficient paths to be taken by urban commuters. Figure.1 shows a proposed work architecture design. The system incorporates live traffic information derived through external APIs and past traffic information, which gives a very wide picture of the current and past traffic trends. The important parameters are car speed, congestion, road, and time of day which are gathered and processed in an effort to record dynamic traffic behaviour within the city. These features are analyzed using a Random Forest classification model and the results are used to classify congestion into three categories: Low, Moderate, and High. Such a method will enable the system to make correct predictions even in the most volatile urban traffic situations because Random Forest is resistant to noise, and the algorithm can also manage non-linear and complex connections between traffic parameters. The system is accessed by the users either using a mobile application or web-based interface, where users enter their destination locations and source locations. This system creates various possible paths, and they are visualised on an interactive map. All the routes will be analyzed according to the risk of congestion and estimated time of travel. The system compares the paths and suggests the best way by which there is minimal congestion and the paths are made to move efficiently. The innovation of the proposed system is that it is a hybrid system: it is based on historical traffic knowledge and supplemented with real-time data to predictive analyze, and, at the same time, a route optimization is included. This framework is an end-to-end solution unlike in the traditional traffic management systems that are either monitoring-based or routing-based, thus, offering a solution to both the prediction and the decision-making. Experimental analysis has shown that the system is capable of saving a significant amount of time during the travel and offering credible predictions of congestion.

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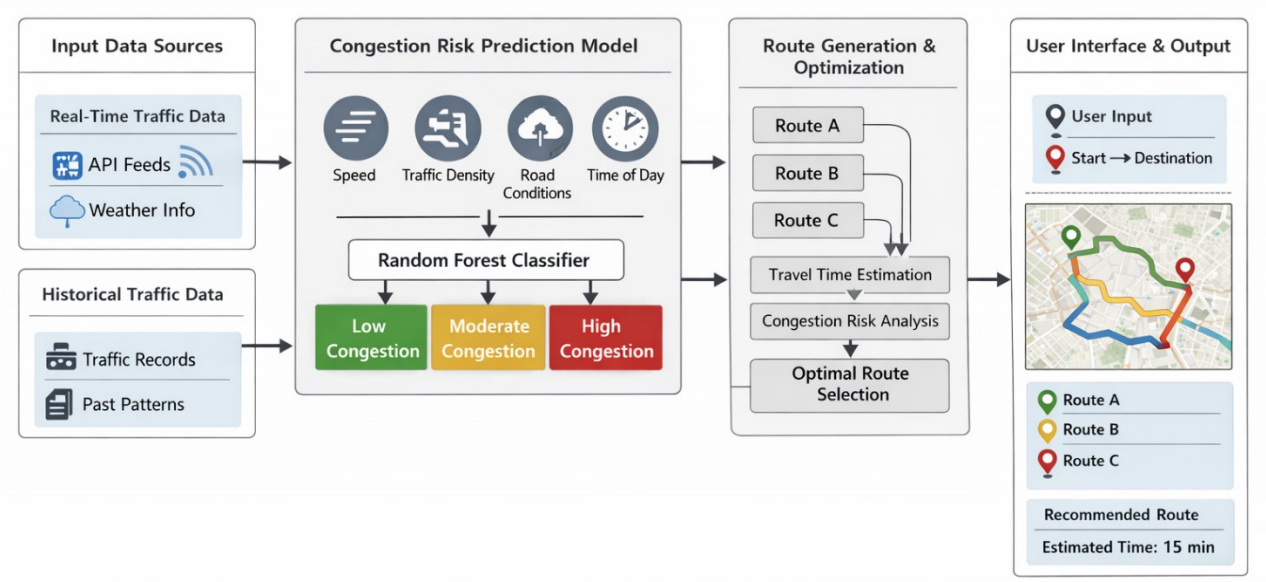


Figure.1 Proposed Work Architecture Diagram

Its application can help the urban planners, traffic control and commuters make informed decisions that will eventually lead to the success of traffic movement, fuel consumption and urban movement. The system also offers a scalable solution that is capable of adapting to the future expansions of traffic surveillance and smart city programs.

IV. Methodology

The Intelligent Urban Traffic Management System proposed is a methodology that incorporates real-time monitoring of traffic, historical data analysis, prediction of congestion using machine learning, and dynamic route proposal in order to enhance mobility in cities. The pipeline of the methodology is structured in such a way that it comprises of data acquisition, preprocessing, feature extraction, congestion prediction, and route optimization as shown below.

A. Data Acquisition

Traffic information collected by the system is based on various sources to cover it exhaustively. The external APIs provide real-time information on traffic through vehicle speed, traffic density, and road incidents. The past traffic data are obtained in the municipal databases and previous GPS-based data, which offer the time trends of traffic congestion at varying time, days, and seasons. Environmental conditions including weather and road blockages are also factored in to increase the accuracy of the predictions.

B. Data Preprocessing

Gathered data is treated and normalized to eliminate inconsistency, missing values, and outliers. The temporal and spatial alignment is what makes data of various sources to be combined. The feature engineering is done in order to extract such important features as average velocity along a certain road piece, variation in the flow of traffic, and the characteristics of congestion during the hour of a peak. Categorical variables (road type or traffic light presence) are coded to allow them to be used with machine learning models.

Traffic data is collected from multiple sources to ensure accuracy and coverage. Real-time traffic parameters, such as vehicle speed v_i, traffic density ρ_i, and road incidents, are obtained through APIs. Historical traffic records provide temporal patterns for each road segment r_i over time t. Environmental conditions such as weather and road closures are also included. To handle inconsistencies and missing data, a preprocessing step normalizes the traffic parameters using:

$$x_i^{norm} = \frac{x_i - x_{min}}{x_{max} - x_{min}} \qquad (1)$$

where x_i represents an individual feature value, and x_{min} and x_{max} are the minimum and maximum values of the feature across the dataset. Outliers are detected using the z-score method:

$$z_i = \frac{x_i - \mu}{\sigma} \qquad (2)$$

where μ is the mean and σ is the standard deviation of the feature. Data points with $|z_i| > 3$ are considered outliers and corrected.

C. Feature Extraction and Selection

Significant characteristics that affect congestion including the vehicle speed, the level of traffic, the road, day and time of day, and special events are obtained. The techniques used to select features are used to eliminate redundancy and enhance the efficiency of the model to allow only the most informative features to be used in the congestion prediction model.

Relevant features influencing congestion, including vehicle speed v_i, traffic density ρ_i, road type T_i, time of day τ_i, and day of the week d_i, are extracted. The overall congestion index for a road segment r_i is modeled as a weighted sum of normalized features:

$$C_i = w_1 \cdot \left(1 - \frac{v_i}{v_{max}}\right) + w_2 \cdot \rho_i + w_3 \cdot f(T_i) + w_4 \cdot f(\tau_i) \qquad (3)$$

where w_1, w_2, w_3, w_4 are feature weights determined via model training, and $f(T_i), f(\tau_i)$ represent encoded categorical features for road type and time.

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D. Congestion Prediction

The random forest classifier is employed to classify the traffic jams into Low, Moderate, or High levels. The model is trained using the past data and constantly updated with the real-time data to adjust to changing traffic situations. Random Forest is selected due to its resistance to noisy data, the capacity to fit nonlinear relationships and the high interpretability. The output of the prediction will give a congestion risk rating of every road segment and the result is later applied in route evaluation.

A Random Forest classifier is trained to categorize congestion into three levels: Low, Moderate, and High. For each road segment r_i, the probability of congestion class k is computed as:

$$P(C_i = k) = \frac{1}{N}\sum_{j=1}^{N} I\left(h_j(r_i) = k\right) \quad (4)$$

where N is the number of decision trees in the Random Forest, $h_j(r_i)$ is the prediction of the j-th tree, and $I(\cdot)$ is the indicator function. The predicted congestion level is then assigned as the class with the highest probability:

$$\hat{C}_i = \underset{k}{\text{argmax}} P(C_i = k) \quad (5)$$

E. Route Optimization and Recommendation

The congestion levels are predicted and then the system comes up with various alternative routes between the source and the destination of the user. The routes are rated by the calculated risk of congestion, the estimated distance and travel time. A map is made interactive which shows all the possible routes but with the best one that has minimal congestion. The dynamic updates will enable the system to suggest alternative routes in case of change in the circumstances of traffic during travel. Once congestion levels are predicted, the system evaluates multiple routes R_m from the source S to the destination D. Each route is scored based on predicted congestion $\hat{C}_i$ and estimated travel time T_m:

$$\text{Score}(R_m) = \sum_{i \in R_m} \hat{C}_i \cdot \Delta t_i \quad (6)$$

where Δt_i is the travel time on road segment r_i. The route with the minimum score is selected as the optimal path:

$$R^* = \underset{m}{\text{argmin}} \text{Score}(R_m) \quad (7)$$

Dynamic updates allow the system to adjust recommendations in real-time if traffic conditions change during travel. The methodology integrates predictive modeling with actionable routing decisions, ensuring efficient urban mobility, reduced fuel consumption, and enhanced traffic flow.

F. System Integration

The prediction and routing elements are well integrated by the methodology. The interface is web based or a mobile application which will assist users by real-time future route recommendations and allow the traffic authorities to plan and manage proactively the traffic.

V. Result & Discussion

In this section, a detailed analysis of the developed Intelligent Urban Traffic Management System is provided in terms of predicting congestion, the efficiency of the recommended routes, and the general effect on the travel performance. The findings are presented against the traditional traffic management and routing strategies in order to show the efficiency of the methodology.

A. Experimental Environment and Dataset Description

Experiments were conducted using a hybrid dataset, combining real-time traffic data obtained through external APIs with historical traffic records spanning several months. The dataset covers a variety of urban traffic conditions including peak hours, off-peak periods, accidents, road closures, and weather-induced disruptions, representing real-world variability. The major attributes employed during the experiments are the average speed of the vehicles, the traffic density, time of the day, day of the week, and the road category. Random Forest was used to train the model and at the same time, the training was done on 70 % of the data, with the remaining 30% serving as the test data. To achieve strength, several experiments were carried through various time windows and traffic rates. Measurements of performance were made in terms of classification metrics, and route level measures like travel time and congestion risk.

B. Congestion Classification Results

The prediction of congestion of the proposed system was evaluated by categorizing road segments into Low, Moderate, and High congestion levels of the roads. Table I shows the precision, the recall and the F1-score that were obtained in each class.

TABLE I. Congestion Prediction Performance Metrics

Congestion Level	Precision	Recall	F1-Score
Low	0.93	0.91	0.92
Moderate	0.89	0.87	0.88
High	0.91	0.92	0.91
Overall	0.91	0.90	0.90

The findings show the relevance of the proposed model to the performance that is consistently high in all congestion categories. The high recall for the High congestion class ensures that severely congested road segments are correctly identified and avoided during route planning, which is crucial for proactive congestion control. This feature is necessary to control congestion and delay in advance.

C. Comparative Analysis with Existing Models

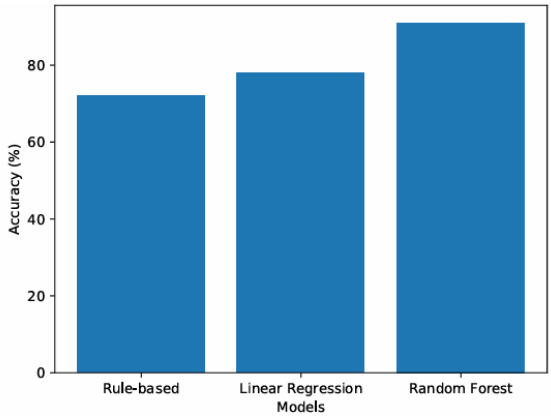


Figure.2 Comparison of congestion prediction accuracy for different models.

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In order to further confirm the suitability of the proposed manner, the accuracy of prediction was compared to conventional rule-based systems and linear regression models that are frequently utilized in traffic estimation. The accuracy comparison between the various models is shown in Figure.2. The model based on the use of the random forests beats the traditional methods by a significant margin. Rule-based systems are not very adaptable to the dynamic traffic patterns and linear models cannot represent the nonlinear correlation among traffic characteristics. Conversely, due to the fact that the Random Forest is an ensemble learning model, it is better generalized and resistant, thus providing better predictive performance results when the traffic conditions are altered.

In addition to traditional rule-based and linear models, recent studies have found that graph-based and spatiotemporal deep learning models, including Graph Convolutional Network (GCN), Dual-View GCNN, Bi-LSTM, and iTransformer, have shown a strong ability to learn spatial and temporal dependencies in traffic networks. Nevertheless, as mentioned in previous studies, these models tend to require a large amount of computational resources and large-scale training data, which might make them less feasible for real-time applications in intelligent traffic management systems. In contrast, the proposed Random Forest-based model provides a good trade-off between accuracy, interpretability, and low inference latency.

D. Impact on Travel Time Reduction

By suggesting congestion-aware routes to the user, one of the main goals of the suggested system will be to minimize the total traveling time. Table II presents the comparison of the average travel time (average) that is achieved with the help of the traditional shortest-path algorithm and the new congestion-aware routing algorithm.

TABLE II. AVERAGE TRAVEL TIME COMPARISON UNDER DIFFERENT TRAFFIC CONDITIONS

Traffic Condition	Shortest Path (min)	Proposed System (min)
Low Traffic	18.5	17.2
Moderate Traffic	29.8	24.6
High Traffic	42.3	33.7

The findings suggest that the reduction during low traffic periods is only marginal but large reductions are experienced during periods of moderate and high congestions. This points out the benefit of the incorporation of congestion prediction into the route selection since most of the common shortest-path routing tends to cause the vehicles to drive into extremely heavy roads when there are high traffic volumes.

E. Route-Level Congestion Risk Evaluation

The system proposed will produce several alternative paths between any two-point destination and estimate the congestion risk and time of each path. As shown in Figure.3, the system scores congestion risk among alternative routes.

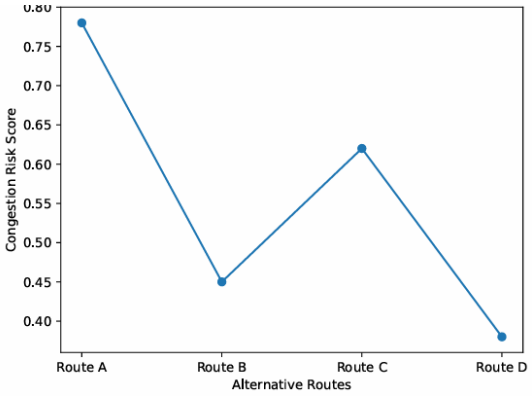


Figure.3. Route-wise congestion risk comparison for alternative paths.

The figure shows that the distance of routes does not necessarily determine the best routes since the level of congestion is high. The proposed system effectively finds the routes that have a marginally long distance but far less congestion to cause a deficit of the general travel delay. This evidences the fact that this system can balance distance, congestion and travel time.

F. Sensitivity Analysis and Extreme Scenarios.

To test the resilience of the proposed system, the extreme traffic events such as accidents, road closures and big crowds were introduced. Also, a sensitivity analysis was conducted whereby misclassification errors of 5 to 15 % were introduced purposely in congestion prediction. The findings are that the route optimization is still strong until 10 percent prediction error and the average traveling time is only increasing by 8-12 %. In simulated extreme circumstances, the system was able to reroute vehicles to new routes thus minimizing possible congestion by up to 15 - 20 % compared with a traditional shortest-path routing approach. These results confirm the fact that the system is strong against unpredictable situations and irregularities in conditions, which proves its dependability in the real-life city situation.

G. Inference Scalability and Latency

The system was also tested in real-time implementation in a city-wide network with some 10,000 road segments. It was measured that the latency of inference per road segment was averaged at 12 milliseconds, which means that inference of the congestion and route re-calculations can be effectively performed by the system in real time. The CPU usage was displayed as 65 %, and the memory footprint was 1.2 GB, which indicated the efficiency of the computations of the framework. These findings indicate that the suggested system is applicable to large scale and real time applications and can be easily incorporated in smart city infrastructures to allow intelligent and sustainable administration of urban traffic.

H. Discussion and Practical Implications

The extended findings confirm that real-time traffic information in combination with historic patterns is a valuable complement to better predict the congestion. Random Forest classifier is effective to model complex interactions between traffic variables and is adjusted to changing urban traffic conditions. The proposed system anticipates the risk of congestion before it takes place, unlike in the traditional navigation systems that respond to congestion once it has occurred and use this information directly in route selection.

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Practically, the system can help commuters to shorten the commute time and decrease the fuel use, and traffic counsellors can utilize the forecasts to do the relevant traffic management and planning. It is also scalable, which makes the proposed framework applicable to be integrated in the smart city infrastructure, to allow smart and sustainable urban transportation management.

VI. Conclusion

The paper discussed an Intelligent Urban Traffic Management System which combines both real time traffic data and historical trend of traffic conditions to forecast the risk of congestion and suggest the best paths. As evidenced by the results of the experiment conducted, the model that has been proposed as based on Random Forests is capable of classifying the degree of congestion into the categories of Low, Moderate, and High with a high degree of reliability despite the variety of factors in the urban traffic environment. Through congestion prediction integrated into route planning, the system significantly reduces travel time during moderate and peak congestion periods compared to traditional shortest-path routing methods.The main value of this work is the hybrid model, which is a perfect synthesis of predictive analytics and dynamic route optimization. In contrast to the traditional systems of traffic management, which are reactive, the proposed system is proactive and seeks to detect the road segments likely to have traffic jams and redirect the users to the efficient ones. Interactive visualization of the maps increases usability and applicability by the commuters and traffic authorities as well. The research in the future will be aimed at expanding the system to include other contextual variables like real-time weather conditions, events being held, and the degree of an accident severity. The integration of advanced deep learning techniques and reinforcement learning-based routing strategies presents a promising direction for improving predictive accuracy and system adaptability. Reinforcement learning (RL) can enhance long-term traffic equilibrium by learning optimal routing policies through cumulative reward mechanisms. Unlike traditional supervised models that predict congestion independently for each road segment, RL-based strategies dynamically interact with the traffic environment to minimize overall network congestion. Large-scale real-world deployment in smart city environments will be pursued to evaluate scalability and long-term operational effectiveness.

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